R-UNIVERSALITY OF AN ODD-DIMENSIONAL SPECIAL ORTHOGONAL LIE ALGEBRA IN TERMS OF THE EVEN CLIFFORD ALGEBRA

DAVE WITTE MORRIS

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Allowed standard prerequisites (editorial).

- The even Clifford algebra of a symmetric matrix over a field, and its Brauer class. Used inside the counted proof at source line 640 with the author's attribution to page 378 of the cited book KnusEtAl-BookOfInvols. Provides the Brauer class that the Tits algebra of the natural representation is equivalent to.
- The remark that in the odd case the centre of the spin group has order two, so that only one Tits algebra has to be considered. The author's remark labelled OnlyZStar, invoked inside the counted proof at source line 639. Reduces the criterion to a single Brauer class, which is what makes the test one-dimensional.

Source reference. Original remark 2.4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 462–462. Navigation pointer only: the counted proof invokes this remark to reduce to a single Tits algebra; its content is declared as a named standard prerequisite.

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Proposition 0.1. *Let* $\mathfrak{g} = \mathfrak{so}_n(B;\mathbb{Q})$*, where* B *is a symmetric matrix in* $\mathrm{GL}_n(\mathbb{Q})$*, and* n *is odd. Then* $\mathfrak{g}$ *is* $\mathbb{R}$*-universal if and only if either* $\mathrm{Cliff}^0_{\mathbb{Q}}(B)$ *is split or* $\mathrm{Cliff}^0_{\mathbb{R}}(B)$ *is not split.*

Proof. Since the center of $\mathbf{Spin}_n$ has order 2 when n is odd, there is only one Tits algebra to consider (see Remark 2.4), and it is Brauer equivalent to $\mathrm{Cliff}^0_F(B)$ [4, p. 378]. □

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